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PRISM*: A Modular Integrated Platform for Health-Sensing Devices

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- ◎ **Real-time Clock**
- ◎ **IMU (6 axis)**
- ◎ **SD Card**
- ◎ **Battery Charging**
- ◎ **MOSFET**
- ◎ **GPIO + IIC**

**PRISM: Participant Recording and Integrated Sensing Module*

INTRODUCTION & BACKGROUND

Key Components For Health Sensing:

Mobile health sensing platforms always require several key components to function effectively [1].

- ***Real-Time Clock (RTC):*** Provides accurate timekeeping
- ***Inertial Measurement Unit (IMU):*** Motion detection
- ***SD Card***: Storage of real-time, timestamped data
- ***PMIC:*** Stable power delivery & battery charging
- ***GPIO & IIC:*** Board Compatiblity & expansion
- ***Mosfet:*** Controls power delivery to peripherals

Limitations On Existing Platform:

However, many existing platforms provide only a subset of these crucial features typically offering three out of four, or even just two—such as lacking either the RTC, IMU, SD card interface—thereby limiting their overall effectiveness.

Xiao Expansion Board (Lacks IMU)

Raspberry Pi 4 (Lacks RTC)

XIAO Grove Shield Extension Board (Lacks SD Card)

METHOD: COMPONENT CHOICE

To ensure our board's compatibility, we incorporated the ***XIAO and QT Py boards' footprints***, which support popular microcontrollers including ***ESP32, RP2040, nRF52840***, etc.

The ***DS3231 RTC clock module*** was selected for its high accuracy and reliable timekeeping, while the ***LSM6DS3 IMU*** was chosen for comprehensive motion sensing including both acceleration and gyroscopic ***6 axis measurements.***

The ***MCP73831*** onboard battery charging circuit supports 3.7-4.2V LiPo batteries with a charging rate of up to ***500mA***.

A ***microSD*** module provides low-power data storage, and ***IIC and GPIO Qwiic connectors*** enable modularity and flexibility for expansion with additional sensors and actuators.

ESP32 with Camera

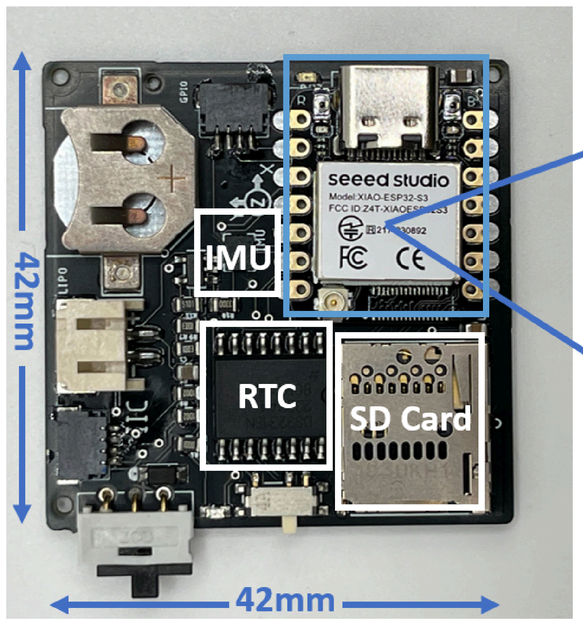
nRF52840

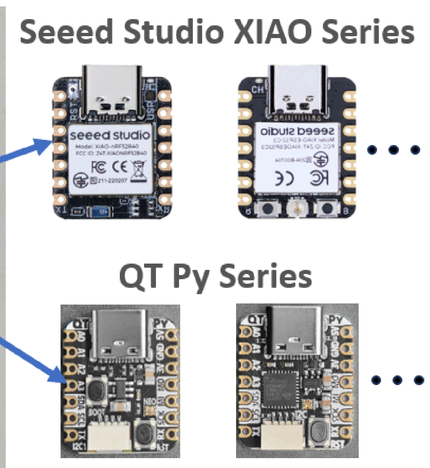
RP2040

RESULTS

A Comprehensive Device for Health-Sensing:

The PRISM platform integrates ***all essential components*** for mobile health sensing on a compact ***42mm by 42mm PCB***, as shown in the Figure. It supports compatibility with 11 different microcontrollers from the XIAO and QT Py series, allowing for flexibility and easy interchangeability in hardware design.





Cost-Effective and Flexible Design for Applicants:

This modular design seamlessly integrates various sensors and peripherals while quickly adapting to diverse project needs. It is currently being used in projects such as ***Eating Gesture Detection and UV Sensing***, making it an ideal solution for mobile health sensing applications. We also prioritize cost control, with each board costing less than ***$14***, and even with microcontroller modules, the total remains under $30—significantly more ***affordable*** than similar development boards.

CONCLUSION & FUTURE WORK

PRISM addresses the limitations of existing mobile health-sensing platforms by integrating essential components such as an RTC for timekeeping, an IMU for motion detection, modular I2C and GPIO connections, SD card storage, and robust power management. This flexible platform simplifies development process and reduces costs. Future work will expand with more sensors and microcontrollers, broadening its applications in diverse fields.

REFERENCE

[1] Y. Yang, H. Wang, R. Jiang, X. Guo, J. Cheng, and Y. Chen, “A review of IoT-enabled mobile healthcare: Technologies,challenges, and future trends,” vol. 9, no. 12, pp. 9478–9502, conference Name: IEEE Internet of Things Journal